Abstract Class

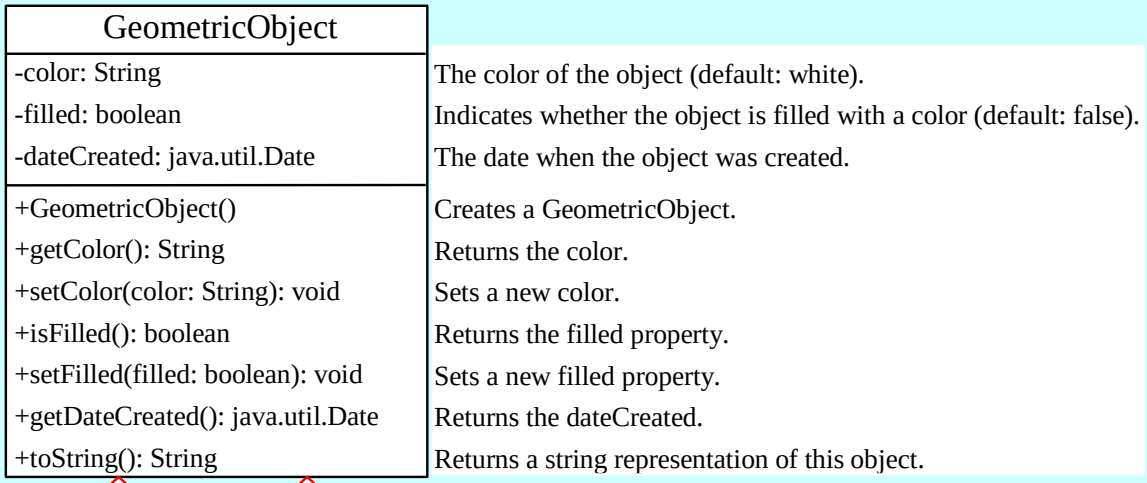
Lecture 21

Based on Slides of Dr. Norazah Yusof

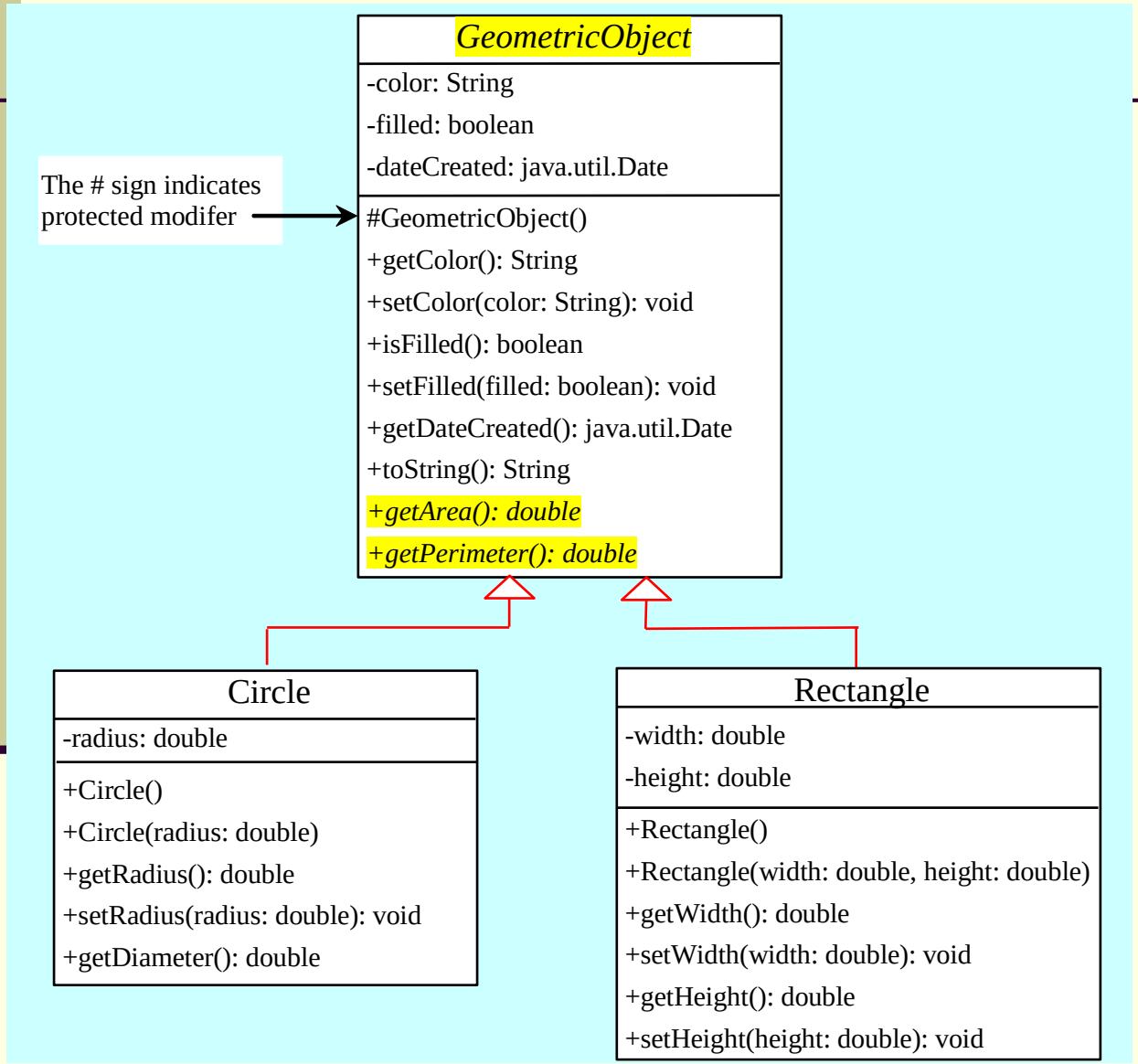
Abstract Class

- **Abstract class is a class with one or more abstract methods.**
- **The `abstract` method**
  - **Method signature without implementation**
- **An abstract class must have a modifier `abstract` included in the class heading.**
- **The `abstract` class**
  - **Cannot be instantiated**
  - **Should be extended and implemented in subclasses**

From Chapter 9



Abstract Classes



Abstract Class

```
public abstract class GeometricObject{
  private String color ="white";
  private boolean filled;
  private java.util.Date dateCreated;

  protected GeometricObject(){
    dateCreated = new java.util.Date();
  }

:
  public abstract double getArea();
  public abstract double getPerimeter();
}
```

Abstract Class

```
public class Circle extends GeometricObject
{
    private double radius;
       :
       :
    public double getArea() {
       return radius * radius * Math.PI;
    }

    public double getPerimeter() {
       return 2 * radius * Math.PI;
    }
}
```

Abstract Class

```
public class Rectangle extends GeometricObject
{
    private double radius;
       :
       :
    public double getArea() {
      return width * height;
 }

    public double getPerimeter() {
       return 2*(width + height);
    }
}
```

NOTE

- **An abstract method cannot be contained in a non-abstract class.**
- **If a subclass of an abstract superclass does not implement all the abstract methods, the subclass must be declared abstract.**
- **In other words, if a non-abstract subclass extended from an abstract class, all the abstract methods must be implemented, even if they are not used in the subclass.**

NOTE

- **An abstract class cannot be instantiated using the new operator, but you can still define its constructors, which are invoked in the constructors of its subclasses.**
- **For instance, the constructors of GeometricObject are invoked in the Circle class and the Rectangle class.**

NOTE

- **A subclass can be abstract even if its superclass is concrete.**
- **For example, the Object class is concrete, but its subclasses, such as GeometricObject, may be abstract.**

NOTE

- **You cannot create an instance from an abstract class using the new operator, but an abstract class can be used as a data type.**
- **Therefore, the following statement, which creates an array whose elements are of GeometricObject type, is correct.**

```
GeometricObject[] geo = new GeometricObject[10];
```

Exercises

- **Self tests**
  - **Question 1, page 174**
  - **Question 2, page 175**
  - **Question 3, page 176**
  - **Question 1, page 192**
- **Lab Exercises:**
  - **Question 2, page 177**
  - **Question 2, page 192**